Meghan Stevens
Math 460
Exam 3

11/20/2025

Problem 1

(a) Step 1: Standardize the age column

$$\text{Standardized age} = \frac{Age - Age_{min}}{Age_{max} - Age_{min}} = \frac{Age - 27}{80 - 27}$$

We have the follows:

| | Age | Sex | Survived |
|---|---|---|---|
| A | 0 | M | 0 |
| B | 0.057 | F | 1 |
| C | 1 | F | 1 |
| D | 0.434 | M | 0 |
| E | 0.623 | F | 0 |
| F | 0.811 | F | 1 |
| G | 0.528 | F | ? |

Step 2: Calculate the distance from G to all the samples.

We have:

$$AG = \sqrt{(0-0.528)^2 + 1^2} = 1.1308$$
$$BG = \sqrt{(0.057-0.528)^2 + 0^2} = 0.471$$
$$CG = \sqrt{(1-0.528)^2 + 0^2} = 0.472$$

$$DG = \sqrt{(0.434-0.528)^2+1^2} = 1.0044$$
$$EG = \sqrt{(0.623-0.528)^2+0^2} = 0.095$$
$$FG = \sqrt{(0.811-0.528)^2+0^2} = 0.283$$

- We have $EG < FG < BG < CG < DG < AG$. E is the nearest neighbor, and E, F, B are the three nearest neighbors.
- Since the nearest neighbor of G is E (EG is the smallest of AG, BG, CG, DG, EG, and FG) thus we predict G has the same outcome as E, which is Not Survived.
- If we use 3NN, the prediction of G would be the majority outcome of the 3 nearest neighbors. The three nearest neighbors of G are E, F, and B. The majority outcomes of E, F, and B is Survived $\left(\frac{0+1+1}{3} = \frac{2}{3} \approx 1\right)$. Thus 3NN instead predicts G Survived. The prediction changes from 1NN (Not Survived) to 3NN (Survived)

(b) Since the distances do not change, the nearest neighbors are still the same as part a.

- For 1NN, the predicted salary of G is the same as E, 10k.
- For 3NN, the predicted salary of G is the average salaries of E, F, and B, which is $\frac{10+100+70}{3} = 60k$.

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Problem 2

(a) The three nearest neighbors are B, A, and C.

- weight = 1

- Predicted Probability of $E = \frac{(1 \cdot Y_B) + (1 \cdot Y_A) + (1 \cdot Y_C)}{3}$

$$= \frac{(1 \cdot 0) + (1 \cdot 1) + (1 \cdot 1)}{3}$$

$$= \frac{0 + 1 + 1}{3} = \boxed{\frac{2}{3}} > \frac{1}{2}$$

$\Rightarrow$ 3NN with distance weighted predict E as x.

- If the probability is greater than or equal to $\frac{1}{2}$, we predict x.

(b) • weight = $\frac{1}{\text{Distance}}$

- Predicted Probability of $E = \frac{\left(\frac{1}{EB} \cdot Y_B\right) + \left(\frac{1}{EA} \cdot Y_A\right) + \left(\frac{1}{EC} \cdot Y_C\right)}{\frac{1}{EB} + \frac{1}{EA} + \frac{1}{EC}}$

$$= \frac{\left(\frac{1}{1.4} \cdot 0\right) + \left(\frac{1}{3} \cdot 1\right) + \left(\frac{1}{3} \cdot 1\right)}{\frac{1}{1.4} + \frac{1}{3} + \frac{1}{3}}$$

$$= \frac{0 + \frac{1}{3} + \frac{1}{3}}{1.380952381}$$

$$= \boxed{0.4827586207} < \frac{1}{2}$$

$$\approx \frac{14}{29}$$

$\Rightarrow$ 3NN with distance weighted predicts E as 0.

ⓒ The four nearest neighbors are B, A, C, and D.

- weight $= \frac{1}{\text{Distance}}$

- Predicted Probability of $E = \frac{\left(\frac{1}{EB} \cdot Y_B\right) + \left(\frac{1}{EA} \cdot Y_A\right) + \left(\frac{1}{EC} \cdot Y_C\right) + \left(\frac{1}{ED} \cdot Y_D\right)}{\frac{1}{EB} + \frac{1}{EA} + \frac{1}{EC} + \frac{1}{ED}}$

$$= \frac{\left(\frac{1}{1.4} \cdot 0\right) + \left(\frac{1}{3} \cdot 1\right) + \left(\frac{1}{3} \cdot 1\right) + \left(\frac{1}{4} \cdot 1\right)}{\frac{1}{1.4} + \frac{1}{3} + \frac{1}{3} + \frac{1}{4}}$$

$\frac{8}{12} + \frac{3}{12}$

$$= \frac{0 + \frac{1}{3} + \frac{1}{3} + \frac{1}{4}}{1.630952381} = \frac{\frac{11}{12}}{1.630952381}$$

$$= \boxed{0.56204} > \frac{1}{2}$$

$\Rightarrow$ 4NN with distance weighted predicts E as x.

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**Problem 3**

(a) We calculate the Manhattan distance from Alice to other users. Let $d_{A1}$ be the Manhattan distance from Alice to User 1 (exclude Item 5). We have:

$$d_{A1} = |5-3| + |3-1| + |3-2| + |4-3| = 6$$

Similarly:

$$d_{A2} = |5-2| + |3-3| + |3-4| + |4-3| = 5$$
$$d_{A3} = |5-3| + |3-3| + |3-1| + |4-4| = 4$$
$$d_{A4} = |5-1| + |3-5| + |3-5| + |4-4| = 8$$

- Thus, the two nearest neighbors are User 3 and User 2, Alice's rating on Item 5 is predicted as the average ratings of these two Users on Item 5, which is $(4+5)/2 = 4.5$.
- We recommend this item to Alice.

(b) We calculate the cosine similarities of Item 5 to other items (excluding Alice). Let $s_{51}$ be the cosine similarity of Item 5 and Item 1. We have:

3 3
3 5
4 4
4 2

$$s_{51} = \frac{3 \cdot 3 + 5 \cdot 2 + 4 \cdot 3 + 2 \cdot 1}{\sqrt{3^2+5^2+4^2+2^2} \cdot \sqrt{3^2+2^2+3^2+1^2}} = \frac{33}{\sqrt{54} \cdot \sqrt{23}} \approx 0.9364$$

Similarly,

$$s_{52} = \frac{3 \cdot 1 + 5 \cdot 3 + 4 \cdot 3 + 2 \cdot 5}{\sqrt{3^2+5^2+4^2+2^2} \cdot \sqrt{1^2+3^2+3^2+5^2}} = \frac{40}{\sqrt{54} \cdot \sqrt{44}} \approx 0.8206$$

$$s_{53} = \frac{3 \cdot 2 + 5 \cdot 4 + 4 \cdot 1 + 2 \cdot 5}{\sqrt{3^2+5^2+4^2+2^2} \cdot \sqrt{2^2+4^2+1^2+5^2}} = \frac{40}{\sqrt{54} \cdot \sqrt{46}} \approx 0.8026$$

$$s_{54} = \frac{3 \cdot 3 + 5 \cdot 3 + 4 \cdot 4 + 2 \cdot 4}{\sqrt{3^2+5^2+4^2+2^2} \cdot \sqrt{3^2+3^2+4^2+4^2}} = \frac{48}{\sqrt{54} \cdot \sqrt{50}} \approx 0.9238$$

- Since $s_{51}$, $s_{54}$, and $s_{52}$ are the largest, the three nearest neighbors (largest similarities) of Item 5 are Item 1, Item 4, and Item 2. Alice's rating on Item 5 is predicted as the average rating of her ratings for Item 1, Item 4, and Item 2, which is $(5+4+3)/3 = 4$.
- We also recommend this item to Alice.